Cosmic Birefringence as Closure-Conditioned Polarisation Transport in Darke Theory

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Abstract

Cosmic birefringence is usually framed as a possible indicator of parity-violating physics beyond the standard electromagnetic and cosmological descriptions. This paper develops a distinct theorem architecture within Darke Theory. The central proposal is that any genuine cosmological polarisation rotation should first be interpreted not as an externally appended exotic interaction, but as a possible observational trace of closure-conditioned propagation in an emergent geometric regime. In this framework geometry is not primitive, admissible propagation exists only in a post-threshold closure sector, and transport need not be exactly parity-neutral if closure transport itself acquires an irreducible handed asymmetry.

The paper has five purposes. First, it states cosmic birefringence in Darke-theoretic terms. Secondly, it defines a parity-sensitive closure field derived from closure transport rather than from a new primitive particle sector. Thirdly, it constructs a line-of-sight transport functional for polarisation rotation. Fourthly, it separates the phenomenon into late-time propagation and fossil-imprint branches. Fifthly, it develops an observational proxy architecture and null-testing framework suitable for testing the proposed structure against CMB birefringence maps and large-scale structure. In addition, the present revision records a first exploratory map-space implementation using masked PR3 component-separated polarisation products. That implementation is explicitly weaker than a full rotation-field analysis, but is strong enough to test whether sign-preserving large-scale directional organisation is present in the cleaned polarisation coherence field.

The central constitutive source is an admissibility-filtered closure-helicity pseudoscalar. A minimal theorem chain is established showing that pure scalar closure-density transport cannot generate birefringence, that anisotropic late-time support induces anisotropic closure response, that anisotropic response generates transverse closure transport, and that non-integrable anisotropic response yields nonzero late-time closure helicity. A further theorem chain establishes that if late-time environment classes induce sign-stable closure-helicity sectors, then exact line-of-sight cancellation is not the natural outcome and a nonzero late-time birefringence field follows on a nonzero-measure set of directions.

The empirical status claimed here remains disciplined. The paper does not claim a completed proof that cosmic birefringence is real or that Darke Theory uniquely explains any present dataset. It does claim something narrower and now more concrete. A masked coarse-cell coherence analysis of Commander and SMICA polarisation maps yields a persistent preferred axis family that converges near $(155°–165°, 48°–50°)$ at intermediate-to-fine coarse resolutions, with corroboration from the 143-GHz foreground-subtracted SEVEM product at the intermediate scale, whereas a more ambitious signed local-transport construction does not yet survive internal shuffle nulls strongly enough to bear explanatory weight by itself. The result is therefore a theorem-level programme with an initial product-persistent anomaly candidate in map-space coherence, not a completed birefringence detection.

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1 Introduction

Cosmic birefringence, broadly understood, refers to a cumulative rotation of the linear polarisation basis of radiation as it propagates across cosmological distance. In standard discussion this possibility is often treated as a potential indicator of parity-violating interactions beyond the ordinary electromagnetic picture, for example through couplings to pseudoscalar fields or other externally appended sectors. The purpose of the present paper is not to review those frameworks, nor to reject them by assertion. Its purpose is to ask a different foundational question:

> If geometry itself is emergent, closure-conditioned, and only post-threshold admissible, what would cosmic birefringence look like when formulated from within that architecture rather than appended to it from outside?

That is the correct entry point inside Darke Theory.
Across the Darke Theory corpus, the relevant structural sequence is

$$\text{recursive differentiation} \Longrightarrow \text{closure} \Longrightarrow \text{admissibility} \Longrightarrow \text{persistent structure.}$$

This sequence is not cosmetic. It reverses the ordinary explanatory order. Space, geometry, objecthood, and lawful propagation are not assumed at the beginning. They are earned downstream of closure and admissibility. The present paper therefore starts from the proposition that if a cosmological polarisation rotation exists, then it should first be read as a possible residual property of the propagation-supporting closure regime itself.

That move matters. It shifts the explanatory burden from "what new field twists light in otherwise neutral spacetime?" to "under what conditions is post-threshold transport itself not perfectly parity-neutral?" The second question belongs naturally inside Darke Theory. The first belongs to a different explanatory order.

The present paper is not a full proof that cosmic birefringence is generated by Darke-theoretic closure structure. Nor does it claim that current data have already settled the issue. Instead it develops a theorem architecture and constitutive programme. The goals are fivefold:

1. to state the phenomenon of cosmic birefringence in DT-native terms;
2. to define a parity-sensitive closure field from admissible closure transport;
3. to construct a line-of-sight transport functional for polarisation rotation;
4. to separate the phenomenon into late-time and fossil-imprint branches;
5. to identify the exact observational and proof burdens required for either branch to be serious.

The paper proceeds in that order. Section 2 introduces notation and status conventions. Section 3 recalls the minimum foundational architecture required from earlier Darke Theory work. Section 4 states the primitive birefringence claim in DT language. Section 5 introduces the parity-sensitive closure field and the closure-helicity pseudoscalar. Section 6 defines the polarisation rotation functional. Section 7 separates the late-time and fossil branches. Section 8 and Section 9 develop the observational proxy and null architecture. Section 10 isolates the internal proof hinges. Section 11 and Section 12 begin the actual proof construction by deriving existence conditions for nonzero closure helicity and the generation of transverse late-time closure transport. Section 13 and Section 14 complete the first constructive chain for the late-time branch by establishing line-of-sight non-cancellation under environment-class coherence. The final sections clarify scope, failure modes, testing outputs, and exact theorem status.

A further clarification is important at the outset. The late-time branch is developed more strongly in the present paper than the fossil branch. This asymmetry is intentional and reflects the current state of the programme rather than a claim of equal completion. The late-time branch admits a sharper constitutive and environment-conditioned theorem chain, whereas the fossil branch is formalised here primarily at the level of branch definition, observational role, and open proof obligation.

The overall posture is deliberately disciplined. The paper does not claim more than it has. But it does claim something substantial: once geometry is treated as closure-conditioned rather than primitive, cosmic birefringence becomes a structurally intelligible problem inside Darke Theory, and one can identify the exact constitutive object on which the full programme rises or falls. The present revision adds a first exploratory implementation note to that architecture. Using masked PR3 component-separated polarisation products already local to the working analysis environment, the programme was pushed far enough to ask a deliberately narrow question: before any full birefringence rotation-field reconstruction is attempted, do the cleaned Q/U fields already exhibit a stable, sign-preserving large-scale coherence asymmetry that survives basic internal controls? The answer is not trivial. A multiscale coarse-cell coherence analysis finds a persistent axis family in Commander and SMICA and a partial match in the 143-GHz foreground-subtracted SEVEM product at the intermediate scale, whereas a more ambitious signed local-transport field fails to survive shuffle nulls strongly enough to be treated

as constitutive evidence. These exploratory results do not settle the birefringence question, but they do sharpen the proof and observational burdens in a way the earlier version could only state abstractly.

2 Notation and status conventions

For clarity, the paper separates four levels of statement.

(1) **Definitions.** These introduce constitutive or observational objects used by the programme.

(2) **Principles.** These state structural commitments internal to the Darke-theoretic framework.

(3) **Derived lemmas and theorems.** These are consequences obtained from the stated constitutive assumptions and previously defined objects.

(4) **Open proof obligations.** These are statements identified as necessary for programme completion but not established in the present paper.

The present paper is a theorem-architecture and constitutive-programme paper. It does not claim that every constitutive assumption has already been derived from a deeper closure-dynamics law, nor that all branch-specific proof obligations have already been discharged. Where structural continuity, nondegeneracy, or admissibility regularity are assumed, that status is stated explicitly.

For convenience, the principal symbols are collected here.

Symbol	Meaning
M	Post-threshold spacetime domain.
Φ	Coherence-density field.
Φ_{crit}	Threshold above which geometry is activated.
J^μ	Closure transport current.
Σ	Closure-balance scalar, $\Sigma = \nabla_\mu J^\mu$.
B, S	Local burden and support densities.
$F(\Phi)$	Geometry-activation factor from the closure-curvature bridge.
$\Omega_{\mu\nu}$	Antisymmetric closure transport tensor, $\Omega_{\mu\nu} = \nabla_\mu J_\nu - \nabla_\nu J_\mu$.
u^μ	Distinguished future-directed timelike transport frame.
H_{cl}	Closure-helicity pseudoscalar, $H_{\mathrm{cl}} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} u_\mu J_\nu \Omega_{\rho\sigma}$.
$\chi(\Phi)$	Threshold activation factor for parity-sensitive transport.
A_{loc}	Local admissibility factor.
Π	Parity-sensitive closure field, $\Pi = A_{\mathrm{loc}}\chi(\Phi)H_{\mathrm{cl}}$.
$W(x)$	Line-of-sight transport weight.
$\gamma(\hat{n})$	Admissible propagation path in sky direction $\hat{n}$.
$\alpha_{\mathrm{DT}}(\hat{n})$	DT birefringence rotation field on the sky.
$\alpha_{\mathrm{late}}, \alpha_{\mathrm{fossil}}$	Late-time and fossil branch contributions.
Q_μ	Transverse closure transport sector beyond pure scalar closure flow.
$\Xi_{\mu\nu}$	Antisymmetric transport content of the transverse sector, $\Xi_{\mu\nu} = \nabla_\mu Q_\nu - \nabla_\nu Q_\mu$.
$\mathcal{K}_\mu{}^\nu$	Closure-response tensor.
$\mathcal{A}_\mu{}^\nu$	Anisotropic part of the closure-response tensor.
$P_{\mathrm{act}}, P_{\mathrm{adm}}, P_{\mathrm{hand}}$	Activation, admissibility, and handedness observational proxies.
$P_{\mathrm{late}}, P_{\mathrm{fossil}}$	Late-time and fossil effective proxy fields.

The paper should therefore be read with the following discipline. Definitions and principles specify the internal architecture. Lemmas and theorems establish consequences within that architecture. Open proof obligations identify the exact places where deeper derivation or empirical confrontation still remains.

3 Foundational setting

Only a small subset of the broader Darke Theory corpus is needed for the present construction. We record the minimum relevant commitments here.

3.1 Recursive differentiation, closure, and admissibility

The foundational work of Darke Theory begins from recursive differentiation rather than from pre-given geometric or material primitives. Stable structure appears only when refinement admits closure, and not every formally describable branch is structurally supportable. Admissibility is therefore the regime filter distinguishing stable carried structure from merely writable structure.

The present paper stands downstream of that general sequence:

$$\text{recursive differentiation} \Longrightarrow \text{closure} \Longrightarrow \text{admissibility} \Longrightarrow \text{persistent structure}.$$

The point is not to rederive those foundations here, but to note the relevant implication. If geometry and propagation appear only downstream of closure, then the default expectation of an exactly parity-neutral propagation background is no longer ontologically primitive. It becomes a condition to be recovered, not assumed.

3.2 Post-threshold geometry

The closure-curvature bridge established the post-threshold regime in which geometry is activated only when the coherence-density field exceeds a critical threshold. The bridge also introduced closure transport, closure balance, support, burden, and admissibility as constitutive parts of the geometric regime. The variables most directly relevant here are:

- the coherence-density field Φ,
- the closure transport current J^μ,
- the closure-balance scalar $\Sigma = \nabla_\mu J^\mu$,
- the burden and support densities,
- the geometry activation factor $F(\Phi)$.

The present paper introduces no new primitive ontology beyond these variables and their lawful constitutive descendants.

3.3 Phase-sensitive observables as residual structure carriers

A repeated empirical pattern across several Darke Theory application papers is that phase-, orientation-, or transport-sensitive observables can retain genuine structural information not exhausted by second-order or amplitude-only summaries. This pattern appears in CMB lensing phase coherence, pre-flare solar phase suppression, turbulence triad conditioning, Kaikōura phase transport, Whakaari phase disorder localisation, and pre-ictal EEG phase structure. The present paper does not assume any one of those results as proof of the present thesis. It does, however, treat them as strong methodological precedent for taking polarisation orientation seriously as a possible carrier of residual structure.

4 Primitive DT statement of cosmic birefringence

We now state the phenomenon in DT-native form.

Definition 4.1 (DT cosmic birefringence)**.** Let M be a post-threshold spacetime domain and let γ be an admissible propagation path for a linearly polarised radiation mode. Cosmic birefringence in Darke Theory is the cumulative rotation of the transported polarisation basis induced by a non-vanishing parity-sensitive asymmetry in the closure-conditioned transport structure sampled along γ.

This definition differs sharply from the most common standard framing. It does not begin by positing a new parity-violating matter sector acting within an otherwise neutral geometric background. It begins instead by asking whether the effective propagation-supporting geometry is itself exactly parity-neutral once one remembers that the geometry is closure-derived.

Principle 4.2 (Propagation non-neutrality principle)**.** If geometry is a post-threshold manifestation of closure rather than an ontologically primitive neutral background, then polarisation transport need not be exactly parity-neutral whenever closure transport carries a persistent handed asymmetry.

The present paper does not claim that such asymmetry must exist. It claims only that once the explanatory order is set by Darke Theory, the possibility is structurally natural and should be investigated at the level of closure transport rather than immediately externalised into an appended exotic field.

5 Constitutive definition of the parity-sensitive closure field

5.1 Why parity-even closure is insufficient

A parity-sensitive propagation effect cannot arise from parity-even scalar closure variables alone.

Lemma 5.1 (No birefringence from parity-even closure alone)**.** *Suppose the post-threshold propagation regime is completely characterised by parity-even scalar quantities Φ, Σ, B, and S, together with a closure transport current that is everywhere orientation-neutral in the relevant propagation sector. Then no parity-sensitive closure source exists and no DT birefringence is generated.*

Proof. The fields Φ, Σ, B, and S are parity-even. By hypothesis the transport current contributes no irreducible handed structure. Any constitutive scalar built from these data remains parity-even. A cumulative polarisation rotation in the present framework requires a parity-sensitive source. Since no such source can be formed, the birefringence source vanishes identically.

This lemma shows that birefringence is not a generic consequence of closure. It requires handed closure transport.

5.2 Closure transport antisymmetry

We therefore introduce the antisymmetric content of closure transport.

Definition 5.2 (Closure transport 2-form)**.** Let J^μ be the closure transport current in the post-threshold regime. Define the antisymmetric closure transport tensor

$$\Omega_{\mu\nu} := \nabla_\mu J_\nu - \nabla_\nu J_\mu.$$

Remark 5.3. If J^μ is globally exact and orientation-neutral in the relevant propagation sector, then $\Omega_{\mu\nu} = 0$. If closure transport contains circulation, handed drift, or orientation-sensitive transport structure, then $\Omega_{\mu\nu} \neq 0$.

Lemma 5.4 (Integrable closure transport gives no antisymmetric sector)**.** *If $J_\mu = \nabla_\mu \psi$ for some smooth scalar ψ, then*

$$\Omega_{\mu\nu} = 0.$$

Proof. Directly,

$$\Omega_{\mu\nu} = \nabla_\mu \nabla_\nu \psi - \nabla_\nu \nabla_\mu \psi = 0$$

on scalar fields.

5.3 Closure-helicity pseudoscalar

To extract a parity-odd scalar source, we contract the antisymmetric transport content with the closure current and the cosmological transport frame.

Definition 5.5 (Closure-helicity pseudoscalar)**.** Let u^μ be a distinguished future-directed timelike unit vector field defining the cosmological transport frame. Define the closure-helicity pseudoscalar

$$H_{\mathrm{cl}} := \epsilon^{\mu\nu\rho\sigma} u_\mu J_\nu \nabla_\rho J_\sigma.$$

Equivalently,

$$H_{\mathrm{cl}} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} u_\mu J_\nu \Omega_{\rho\sigma}.$$

Remark 5.6. H_{cl} is not matter helicity, electromagnetic helicity, or an inserted new field. It is a constitutive measure of the parity-odd content of closure transport itself.

Lemma 5.7 (Vanishing under parity-neutral transport)**.** *If closure transport is orientation-neutral in the relevant propagation sector, then*

$$H_{\mathrm{cl}} = 0.$$

Proof. Orientation-neutral transport contains no parity-odd contraction between u^μ, J^μ, and the antisymmetric transport sector. Hence the Levi-Civita contraction vanishes.

Lemma 5.8 (Vanishing under globally integrable transport)**.** *If J^μ is globally integrable on the admissible propagation domain, then*

$$H_{\mathrm{cl}} = 0.$$

Proof. By the previous integrability lemma, $\Omega_{\mu\nu} = 0$. Substituting into the equivalent form of H_{cl} gives the claim.

5.4 Threshold activation and admissibility filtering

Since geometry exists only in the post-threshold regime, the parity-sensitive source must be gated by threshold activation and local admissibility.

Definition 5.9 (Threshold activation factor)**.** Let

$$\chi(\Phi)$$

be a smooth non-negative function satisfying

$$\chi(\Phi) = 0 \quad \text{for } \Phi \le \Phi_{\mathrm{crit}}, \qquad \chi(\Phi) > 0 \quad \text{for } \Phi > \Phi_{\mathrm{crit}}.$$

Definition 5.10 (Local admissibility factor)**.** Let

$$A_{\mathrm{loc}}(x) = \mathcal{A}(S(x), B(x), \Sigma(x), \Phi(x))$$

be a bounded local admissibility factor satisfying

$$0 \le A_{\mathrm{loc}}(x) \le 1,$$

with A_{loc} increasing with local support and decreasing with local burden.

Definition 5.11 (Parity-sensitive closure field)**.** Define the parity-sensitive closure field by

$$\Pi(x) := A_{\mathrm{loc}}(x)\,\chi(\Phi(x))\,H_{\mathrm{cl}}(x).$$

This is the minimal constitutive DT source for birefringence.

6 Polarisation transport functional

We now define the observable transport law.

Definition 6.1 (Pathwise rotation angle)**.** Let γ be an admissible propagation path parameterised by λ. Let $W(x) \geq 0$ be a transport weight encoding propagation visibility and support. Define the DT polarisation rotation angle along γ by

$$\alpha_{\mathrm{DT}}(\gamma) = \int_\gamma W(x)\,\Pi(x)\,d\lambda.$$

Equivalently,

$$\alpha_{\mathrm{DT}}(\gamma) = \int_\gamma W(x)\,A_{\mathrm{loc}}(x)\,\chi(\Phi(x))\,H_{\mathrm{cl}}(x)\,d\lambda.$$

Definition 6.2 (Observed sky rotation field)**.** For line-of-sight direction $\hat{n}$, let $\gamma(\hat{n})$ denote the corresponding admissible path from the effective last-scattering surface to the observer. Define

$$\alpha_{\mathrm{DT}}(\hat{n}) = \int_{\gamma(\hat{n})} W(x)\,\Pi(x)\,d\lambda.$$

Lemma 6.3 (Vanishing under vanishing parity-sensitive closure field)**.** *If $\Pi(x) = 0$ along every admissible path $\gamma(\hat{n})$, then*

$$\alpha_{\mathrm{DT}}(\hat{n}) = 0$$

for every $\hat{n}$.

Proof. The integrand vanishes pointwise, so the line integral vanishes.

Lemma 6.4 (Finite cumulative rotation under bounded sign-coherent source)**.** *Suppose Π is bounded and sign-coherent over a subsegment of $\gamma(\hat{n})$ with positive weighted measure. Then*

$$\alpha_{\mathrm{DT}}(\hat{n}) \neq 0.$$

Proof. On such a subsegment the integrand has fixed nonzero sign and positive weighted measure, so its integral is nonzero. Therefore the full path integral is nonzero.

Principle 6.5 (Residual-orientation principle)**.** A weak but sign-coherent parity-sensitive closure source can generate a detectable rotation in polarisation-orientation observables while leaving amplitude-based power summaries only weakly affected.

7 Late-time and fossil branches

The constitutive source permits two distinct physical realisations.

7.1 Late-time propagation branch

Definition 7.1 (Late-time branch)**.** The late-time branch is the regime in which the parity-sensitive closure field is generated predominantly by evolved large-scale structure after recombination:

$$\Pi(x) = \Pi_{\text{late}}(x),$$

so that

$$\alpha_{\text{late}}(\hat{n}) = \int_{\gamma(\hat{n})} W(x)\, \Pi_{\text{late}}(x)\, d\lambda.$$

In constitutive form,

$$\Pi_{\text{late}}(x) = A_{\text{loc}}(x)\, \chi(\Phi(x))\, H_{\text{late}}(x),$$

where H_{late} is the late-time closure-helicity density generated by evolved large-scale structure.

7.2 Fossil-imprint branch

Definition 7.2 (Fossil branch)**.** The fossil branch is the regime in which a parity-sensitive closure asymmetry is generated during or near the geometry-activation era and survives as a relic imprint in the later propagation sector. In that case one writes

$$\alpha_{\text{DT}}(\hat{n}) = \alpha_{\text{fossil}}(\hat{n}) + \int_{\gamma(\hat{n})} W(x)\, \Pi_{\text{late}}(x)\, d\lambda.$$

Remark 7.3. The fossil contribution may be interpreted as a boundary or initial-condition term inherited from early closure activation.

Proposition 7.4 (Mixed-branch decomposition)**.** *The observed DT birefringence field may in general be decomposed as*

$$\alpha_{\text{DT}}(\hat{n}) = \alpha_{\text{fossil}}(\hat{n}) + \alpha_{\text{late}}(\hat{n}),$$

with one branch possibly dominating observationally.

8 Observational proxy architecture

The constitutive field $\Pi = A_{\text{loc}}\chi(\Phi)H_{\text{cl}}$ is not directly observable. The next task is therefore to identify measurable sky fields that should track the integrated DT source if the constitutive picture is correct.

8.1 Proxy principle

Principle 8.1 (Proxy principle)**.** No scalar density field alone can serve as a complete observational proxy for DT birefringence. A valid proxy architecture must preserve at least one of the following:

(a) handedness or sign structure,

(b) line-of-sight asymmetry,

(c) environment-conditioned transport anisotropy,

(d) residual phase-sensitive organisation.

Definition 8.2 (Observational proxy field)**.** An observational proxy field for DT birefringence is any measurable sky field $P(\hat{n})$ such that variation in $P(\hat{n})$ is expected, under the DT constitutive model, to track variation in the line-of-sight integral

$$\int_{\gamma(\hat{n})} W\, A_{\text{loc}}\, \chi(\Phi)\, H_{\text{cl}}\, d\lambda$$

up to smoothing, projection, or monotone distortion.

8.2 Proxy classes

The constitutive form suggests three broad classes of proxy.

Definition 8.3 (Activation proxy)**.** An activation proxy $P_{\mathrm{act}}(\hat{n})$ is a sky field intended to track where post-threshold geometric support and persistent large-scale structure are strongest along the line of sight.

Definition 8.4 (Admissibility proxy)**.** An admissibility proxy $P_{\mathrm{adm}}(\hat{n})$ is a sky field intended to track whether the sampled line of sight lies in a structurally coherent, supportable, non-random propagation sector.

Definition 8.5 (Handedness proxy)**.** A handedness proxy $P_{\mathrm{hand}}(\hat{n})$ is a signed sky field intended to track the sign and magnitude of line-of-sight handed closure transport.

Definition 8.6 (Factorised effective proxy)**.** Define the factorised effective proxy field by

$$P_{\mathrm{eff}}(\hat{n}) := P_{\mathrm{act}}(\hat{n})\, P_{\mathrm{adm}}(\hat{n})\, P_{\mathrm{hand}}(\hat{n}).$$

Principle 8.7 (Factorisation principle)**.** If the DT constitutive model is correct, then a combined proxy preserving activation, admissibility, and handedness should outperform parity-even scalar environment fields that preserve only density or only activation.

8.3 Late-time proxy architecture

For the late-time branch, define

$$P_{\mathrm{late}}(\hat{n}) := P^{\mathrm{late}}_{\mathrm{act}}(\hat{n})\, P^{\mathrm{late}}_{\mathrm{adm}}(\hat{n})\, P^{\mathrm{late}}_{\mathrm{hand}}(\hat{n}).$$

The exact catalogue-level implementation is deferred to later work. At theorem-architecture stage, it is sufficient that $P^{\mathrm{late}}_{\mathrm{act}}$ encode line-of-sight structural depth, $P^{\mathrm{late}}_{\mathrm{adm}}$ encode environment-conditioned supportability, and $P^{\mathrm{late}}_{\mathrm{hand}}$ encode sign-preserving asymmetry rather than scalar density alone.

8.4 Fossil proxy architecture

For the fossil branch, define

$$P_{\mathrm{fossil}}(\hat{n}) := P_{\mathrm{prim}}(\hat{n})\, P_{\mathrm{low}\ell}(\hat{n})\, P_{\mathrm{par}}(\hat{n}),$$

where these fields are intended to encode broad parity-sensitive primordial or low-ℓ structure rather than late-time environment.

9 Minimal statistical programme and null architecture

The proxy architecture is not enough. One must specify what counts as support, what nulls are required, and what would count against the programme.

9.1 Primary observational relation

Definition 9.1 (Primary observational relation)**.** The primary observational relation is the structured dependence

$$\alpha_{\mathrm{obs}}(\hat{n}) \sim \mathcal{F}\big(P_{\mathrm{act}}(\hat{n}), P_{\mathrm{adm}}(\hat{n}), P_{\mathrm{hand}}(\hat{n})\big) + \epsilon(\hat{n}),$$

where ϵ denotes instrumental, foreground, and unresolved residual contributions.

At minimal level $\mathcal{F}$ may be taken to be linear. At more refined level it may be conditional, thresholded, or sign-sensitive.

9.2 Hierarchy of tests

The DT statistical programme should proceed in the following order.

Stage I. Scalar-vs-structural comparison. Compare signed structural proxies against parity-even scalar density proxies.

Stage II. Environment-conditioning test. Test whether conditioning on line-of-sight environment materially increases explanatory power.

Stage III. Sign-preservation test. Test whether destroying handed sign structure collapses performance.

Stage IV. Branch discrimination. Only then compare fossil, late-time, and mixed models.

9.3 Performance functionals

Definition 9.2 (Performance functional)**.** A performance functional $\mathrm{Perf}(\alpha, P)$ is any statistic measuring structured dependence between an observed rotation field $\alpha(\hat{n})$ and a proxy field $P(\hat{n})$.

Acceptable choices include signed correlation, rank correlation, regression coefficient stability, conditional mean separation, and likelihood improvement relative to null templates.

Principle 9.3 (Sign-sensitivity requirement)**.** At least one primary performance functional must preserve sign information. Purely magnitude-based comparisons are insufficient for testing a parity-sensitive closure-helicity model.

Definition 9.4 (Sector contrast)**.** For a signed proxy $P(\hat{n})$, define high-sign and low-sign sectors

$$S_+ := \{\hat{n} : P(\hat{n}) > q_+\}, \qquad S_- := \{\hat{n} : P(\hat{n}) < q_-\},$$

where q_+ and q_- are high and low quantile thresholds. Define the sector contrast

$$\Delta_{\mathrm{sec}}(\alpha; P) := \langle \alpha \rangle_{S_+} - \langle \alpha \rangle_{S_-}.$$

9.4 Minimal null architecture

A serious DT birefringence test requires multiple nulls.

Definition 9.5 (Scalar-preserving sign-destruction null)**.** Construct a null proxy field by preserving the magnitude and spatial support of the handedness proxy while scrambling or randomising its sign structure.

Definition 9.6 (Environment-preserving handedness-destruction null)**.** Construct a null

$$P^{(2)}_{\mathrm{null}} = P_{\mathrm{act}} P_{\mathrm{adm}} \widetilde{P}_{\mathrm{hand}},$$

where $\widetilde{P}_{\mathrm{hand}}$ preserves broad smoothing scale and marginal distribution but destroys coherent sign alignment with environment.

Definition 9.7 (Sky-remapping null)**.** Construct rotated or remapped proxy fields $P_R(\hat{n})$ preserving broad field statistics while breaking true sky alignment.

Definition 9.8 (Random-sector null)**.** Construct random sky sectors matched in size, mask, and smoothness to proxy-defined sectors and compare observed sector contrasts against the random ensemble.

Definition 9.9 (Frequency-instability null)**.** Compare the inferred relation between α_{obs} and the DT proxies across observing bands after foreground cleaning. Strong ordinary frequency dependence weakens the structural-geometric interpretation.

9.5 Support conditions for the late-time branch

The late-time branch receives observational support only if all of the following hold simultaneously:

(1) a signed structural proxy outperforms parity-even density proxies;

(2) environment conditioning materially increases explanatory power;

(3) sign-scramble nulls substantially reduce performance;

(4) sky-remapping nulls weaken the association;

(5) the result is not strongly frequency-unstable.

9.6 Support conditions for the fossil branch

The fossil branch receives observational support if

(1) late-time environment conditioning yields little gain;

(2) broad parity-sensitive large-angle proxies retain explanatory power;

(3) the signal is approximately frequency-stable;

(4) low-ℓ or primordial parity structure matters more than local environment segmentation.

9.7 Exploratory map-space implementation note

The initial implementation carried out for the present revision does not yet start from an externally reconstructed birefringence field $\alpha_{\rm obs}(\hat{n})$. Instead it tests whether the cleaned polarisation maps themselves already contain product-persistent large-scale directional organisation of the kind that a later signed proxy or rotation-field analysis would need in order to be interesting. The operational object is a masked coarse-cell coherence statistic built from local means of Q and U over nested HEALPix parent cells after strict use of the common polarisation mask.

At this exploratory stage the key logic is as follows.

(1) Build coarse-cell coherence fields from masked Q/U data at several coarse resolutions.

(2) Scan for the preferred axis of sign-preserving hemispheric asymmetry in that coherence field.

(3) Test robustness under tail jackknife and shuffle nulls.

(4) Compare the preferred axis family across independently cleaned products.

This implementation is weaker than the full statistical programme above in three ways. First, it is map-space rather than rotation-field based. Secondly, it does not yet implement the full late-time proxy factorisation $P_{\rm act}P_{\rm adm}P_{\rm hand}$. Thirdly, it cannot yet distinguish a true birefringence rotation from any other mechanism that leaves a persistent sign-preserving organisation in cleaned large-angle polarisation structure. Nevertheless, the exercise is useful because it immediately tests whether the data products contain only unstable noise-like organisation or whether there is a reproducible axis family to explain.

The outcome of the exploratory implementation should therefore be interpreted according to the following rule: a stable product-persistent coherence axis strengthens the case for proceeding to a full DT birefringence test, whereas the absence of such persistence would weaken the motivation for the present constitutive route before any heavier reconstruction effort is undertaken.

9.8 Failure conditions

The framework is weakened or fails under any of the following conditions:

(1) parity-even scalar density proxies perform as well as or better than all signed structural proxies;

(2) sign-scrambling leaves performance essentially unchanged;

(3) sky-remapping nulls preserve the association at similar strength;

(4) the apparent signal is strongly frequency-unstable in a way characteristic of foreground or ordinary propagation contamination;

(5) neither late-time nor fossil proxy architectures outperform generic broad-sky nuisance structure.

10 Branch-specific proof obligations and internal load-bearing hinges

The paper architecture is now complete enough that the remaining burdens can be stated sharply.

10.1 Universal proof hinges

There are four universal hinges.

(1) **Existence of nonzero closure helicity.** One must show that admissible post-threshold closure transport can possess $H_{\mathrm{cl}} \neq 0$.

(2) **Admissibility of parity-sensitive transport.** One must show that parity-sensitive closure transport is not universally filtered out by support, burden, or closure balance.

(3) **Cumulative sign coherence.** One must show that line-of-sight cancellation is not generic.

(4) **Compatibility with a weak observational imprint.** One must show that the effect can live naturally in a residual orientation sector without grossly disrupting the successful large-scale radiative structure.

10.2 Late-time proof obligations

The late-time branch requires:

(1) a generation theorem showing that evolved large-scale structure induces transverse closure transport;

(2) an environment-imprint theorem showing that the resulting birefringence is direction-dependent rather than purely isotropic;

(3) a scalar-insufficiency theorem showing that parity-even scalar environment structure alone cannot reproduce the constitutive effect.

10.3 Fossil proof obligations

The fossil branch requires:

(1) an activation-asymmetry theorem showing that geometry activation is not forced to be exactly parity-neutral;

(2) an imprint-persistence theorem showing that a parity-sensitive residue can survive later propagation;

(3) a weak-environment-coupling theorem distinguishing the fossil branch from the late-time branch.

10.4 Mixed-model obligations

The mixed model requires:

(1) a non-degeneracy theorem showing that the fossil and late-time branches are structurally distinct;

(2) an identifiability theorem showing that one can in principle separate them observationally.

11 Existence conditions for nonzero closure helicity

We now begin the actual proof construction. The first decisive question is:

$$H_{\mathrm{cl}} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} u_\mu J_\nu \Omega_{\rho\sigma},$$

can this ever be nonzero in an admissible DT regime, or is it always forced to vanish?

11.1 No parity-neutrality theorem is built into the current bridge

Lemma 11.1 (No parity-neutrality theorem in the current DT bridge)**.** *The current DT bridge architecture imposes threshold activation, admissibility, and closure balance, but does not contain any theorem requiring every admissible closure transport current J^μ to be globally parity-neutral.*

Proof. The bridge introduces J^μ, Σ, burden B, support S, and activation $F(\Phi)$, but nowhere states or implies that all admissible transport currents are exact gradients with vanishing antisymmetric transport content. Hence exact parity-neutrality is not implied by the current bridge architecture.

11.2 Pure scalar closure transport is too restrictive

The minimal closure-current ansatz from the bridge was

$$J_\mu = \zeta(\Phi)\nabla_\mu \Phi.$$

This is sufficient for the original bridge theorem chain, but too restrictive if treated as globally exhaustive for birefringence.

Lemma 11.2 (Scalar-current vanishing lemma)**.** *If the closure transport current is globally of the pure scalar form*

$$J_\mu = \zeta(\Phi)\nabla_\mu \Phi$$

with ζ depending only on Φ, then

$$\Omega_{\mu\nu} = 0 \qquad \textit{and} \qquad H_{\mathrm{cl}} = 0.$$

Proof. Differentiate:

$$\Omega_{\mu\nu} = \nabla_\mu(\zeta\nabla_\nu\Phi) - \nabla_\nu(\zeta\nabla_\mu\Phi) = (\nabla_\mu\zeta)(\nabla_\nu\Phi) - (\nabla_\nu\zeta)(\nabla_\mu\Phi).$$

Since $\zeta = \zeta(\Phi)$, one has $\nabla_\mu\zeta = \zeta'(\Phi)\nabla_\mu\Phi$. Therefore

$$\Omega_{\mu\nu} = \zeta'(\Phi)\big(\nabla_\mu\Phi\nabla_\nu\Phi - \nabla_\nu\Phi\nabla_\mu\Phi\big) = 0.$$

Substituting into the definition of H_{cl} yields $H_{\mathrm{cl}} = 0$.

This lemma isolates the exact problem. To obtain birefringence, closure transport must contain more than pure scalar closure-density flow.

11.3 Minimal extension: transverse closure transport

We therefore introduce the weakest DT-consistent extension.

Definition 11.3 (Extended closure transport decomposition)**.** Write the closure transport current as

$$J_\mu = J_\mu^{(0)} + Q_\mu,$$

where

$$J_\mu^{(0)} := \zeta(\Phi)\nabla_\mu\Phi$$

is the scalar closure-density flow and Q_μ is a transverse structural correction.

The field Q_μ is not introduced as a new primitive. It is interpreted as anisotropic, circulation-like, or memory-bearing closure transport not reducible to scalar density flow alone.

Definition 11.4 (Antisymmetric transport content of the transverse sector)**.** Define

$$\Xi_{\mu\nu} := \nabla_\mu Q_\nu - \nabla_\nu Q_\mu.$$

Proposition 11.5 (Minimal nonzero-helicity criterion)**.** *Let*

$$J_\mu = \zeta(\Phi)\nabla_\mu\Phi + Q_\mu.$$

If $Q_\mu \neq 0$ on some post-threshold domain, $\Xi_{\mu\nu} \neq 0$ there, and the triple $(u^\mu, J^\mu, \Xi_{\rho\sigma})$ is not everywhere Levi-Civita-degenerate, then

$$H_{\mathrm{cl}} \neq 0$$

on a nonempty subset of the domain.

Proof. The scalar sector contributes no antisymmetric transport content by the scalar-current lemma. Hence $\Omega_{\mu\nu} = \Xi_{\mu\nu}$. Substituting into

$$H_{\mathrm{cl}} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}u_\mu J_\nu \Omega_{\rho\sigma}$$

gives

$$H_{\mathrm{cl}} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}u_\mu J_\nu \Xi_{\rho\sigma}.$$

If $Q_\mu \neq 0$, $\Xi_{\mu\nu} \neq 0$, and the Levi-Civita contraction is not everywhere degenerate, then the pseudoscalar is nonzero on a nonempty subset.

Definition 11.6 (Admissible transverse closure transport)**.** A transverse closure transport sector Q_μ is admissible on a post-threshold domain if the total current J_μ remains smooth, the induced burden remains finite, support remains positive, closure balance does not force immediate suppression, and the activation condition remains satisfied.

Lemma 11.7 (No automatic inadmissibility of transverse transport)**.** *The existence of a transverse closure transport sector Q_μ is not automatically ruled out by the current DT burden-support framework.*

Proof. The current framework constrains whether a regime can persist, but contains no theorem excluding every bounded non-integrable perturbation of the closure transport current. Hence transverse transport is not universally forbidden.

Theorem 11.8 (Existence theorem for nonzero closure helicity)**.** *Let D be a post-threshold domain with scalar closure current*

$$J_\mu^{(0)} = \zeta(\Phi)\nabla_\mu\Phi$$

and an admissible transverse sector Q_μ such that

(1) $Q_\mu \neq 0$ on a nonempty subset of D,

(2) $\Xi_{\mu\nu} \neq 0$ on a nonempty subset of D,

(3) the triple $(u^\mu, J^\mu, \Xi_{\rho\sigma})$ is not everywhere Levi-Civita-degenerate.

Then

$$H_{\mathrm{cl}} \neq 0$$

on a nonempty subset of D.

Proof. By the scalar-current vanishing lemma, the scalar sector contributes no antisymmetric transport content. Hence $\Omega_{\mu\nu} = \Xi_{\mu\nu}$. The previous proposition then applies directly.

12 Late-time generation of transverse closure transport

We now turn to the first branch-specific proof construction. The goal is to show that late-time evolved large-scale structure naturally generates an admissible transverse closure transport sector.

12.1 The scalar-current obstruction

The previous section showed that pure scalar closure-density transport cannot generate birefringence. We make one further point explicit.

Lemma 12.1 (Scalar-current transport limitation)**.** *If*

$$J_\mu = \zeta(\Phi)\nabla_\mu\Phi,$$

then transport is everywhere aligned with the local closure-density gradient and carries no independent transverse transport information.

Proof. The current at each point is proportional to $\nabla_\mu\Phi$. Therefore its direction is fully determined by the local scalar closure-density gradient, leaving no independent transverse transport degree of freedom.

This is the exact reason the late-time branch must produce more than a scalar response.

12.2 Environment-conditioned closure transport

Definition 12.2 (Environment-conditioned closure transport)**.** Closure transport is environment-conditioned if the effective transport response at a point depends not only on the local value of Φ, but also on anisotropic relational or structural context not reducible to Φ alone.

The relevant contexts include directional support anisotropy, closure-memory skew, and large-scale line-of-sight asymmetry.

Definition 12.3 (Anisotropic closure-response tensor)**.** Let

$$\mathcal{K}_\mu{}^\nu$$

be a closure-response tensor field and define

$$J_\mu = \mathcal{K}_\mu{}^\nu \nabla_\nu \Phi.$$

If

$$\mathcal{K}_\mu{}^\nu = \zeta(\Phi)\delta_\mu{}^\nu,$$

one recovers the scalar sector.

Lemma 12.4 (Anisotropic support lemma)**.** *If the admissible support landscape of the late-time post-threshold medium is direction-dependent, then the closure-response tensor $\mathcal{K}_\mu{}^\nu$ is not, in general, everywhere proportional to $\delta_\mu{}^\nu$.*

Proof. A tensor proportional to $\delta_\mu{}^\nu$ weights all directions identically up to a scalar factor. A genuinely direction-dependent support landscape cannot be represented in that way except in the trivial case where the directional dependence vanishes. Hence a nontrivial anisotropic support landscape requires

$$\mathcal{K}_\mu{}^\nu \not\propto \delta_\mu{}^\nu.$$

Corollary 12.5. *If*

$$\mathcal{K}_\mu{}^\nu \not\propto \delta_\mu{}^\nu,$$

then the closure current need not be everywhere parallel to $\nabla_\mu \Phi$, and a transverse sector Q_μ is present.

Proof. Write

$$\mathcal{K}_\mu{}^\nu = \zeta(\Phi)\delta_\mu{}^\nu + \mathcal{A}_\mu{}^\nu,$$

where $\mathcal{A}_\mu{}^\nu$ is the anisotropic part. Then

$$J_\mu = \zeta(\Phi)\nabla_\mu \Phi + \mathcal{A}_\mu{}^\nu \nabla_\nu \Phi.$$

Define

$$Q_\mu := \mathcal{A}_\mu{}^\nu \nabla_\nu \Phi.$$

If $\mathcal{A}_\mu{}^\nu \neq 0$, then $Q_\mu \neq 0$ wherever $\nabla_\nu \Phi$ has nontrivial support.

Principle 12.6 (Late-time constitutive anisotropy principle)**.** If late-time cosmic structure is environment-conditioned in a way not reducible to a single isotropic scalar field, then the closure transport response is anisotropic.

Theorem 12.7 (Late-time generation theorem for transverse closure transport)**.** *Let D be a post-threshold late-time cosmological domain with coherence-density field Φ. Suppose*

(1) the local admissible support structure is direction-dependent on a nonempty subset of D,

(2) closure transport responds constitutively to that support structure,

(3) the response is encoded by a closure-response tensor $\mathcal{K}_\mu{}^\nu$.

Then on that subset the closure transport current is not reducible to the pure scalar form

$$J_\mu = \zeta(\Phi)\nabla_\mu\Phi.$$

Equivalently, there exists a nontrivial transverse closure transport sector

$$Q_\mu = (\mathcal{K}_\mu{}^\nu - \zeta(\Phi)\delta_\mu{}^\nu)\,\nabla_\nu\Phi.$$

Proof. By the anisotropic support lemma,

$$\mathcal{K}_\mu{}^\nu \not\propto \delta_\mu{}^\nu$$

on the relevant subset. Decompose

$$\mathcal{K}_\mu{}^\nu = \zeta(\Phi)\delta_\mu{}^\nu + \mathcal{A}_\mu{}^\nu,$$

with $\mathcal{A}_\mu{}^\nu \neq 0$. Then

$$J_\mu = \zeta(\Phi)\nabla_\mu\Phi + \mathcal{A}_\mu{}^\nu\nabla_\nu\Phi = J_\mu^{(0)} + Q_\mu.$$

Since $\mathcal{A}_\mu{}^\nu \neq 0$, the transverse sector is nontrivial wherever $\nabla_\nu\Phi$ has nontrivial support.

12.3 From transverse transport to nonzero late-time helicity

Lemma 12.8 (Anisotropic non-integrability lemma)**.** *Let*

$$Q_\mu = \mathcal{A}_\mu{}^\nu\nabla_\nu\Phi.$$

If $\mathcal{A}_\mu{}^\nu$ varies directionally across the domain in a way not reducible to a scalar multiple of a gradient-potential structure, then

$$\Xi_{\mu\nu} := \nabla_\mu Q_\nu - \nabla_\nu Q_\mu \neq 0$$

on a nonempty subset.

Proof. Differentiate:

$$\nabla_\mu Q_\nu = (\nabla_\mu\mathcal{A}_\nu{}^\lambda)\nabla_\lambda\Phi + \mathcal{A}_\nu{}^\lambda\nabla_\mu\nabla_\lambda\Phi.$$

Antisymmetrising gives

$$\Xi_{\mu\nu} = (\nabla_\mu\mathcal{A}_\nu{}^\lambda - \nabla_\nu\mathcal{A}_\mu{}^\lambda)\nabla_\lambda\Phi + \mathcal{A}_\nu{}^\lambda\nabla_\mu\nabla_\lambda\Phi - \mathcal{A}_\mu{}^\lambda\nabla_\nu\nabla_\lambda\Phi.$$

If $\mathcal{A}_\mu{}^\nu$ is directionally varying and not reducible to a scalar potential structure, the antisymmetric combination is not forced to vanish. Hence $\Xi_{\mu\nu} \neq 0$ on a nonempty subset.

Theorem 12.9 (Late-time nonzero-helicity theorem)**.** *Let D be a post-threshold late-time cosmological domain. Suppose*

(1) the admissible support structure is direction-dependent on a nonempty subset of D,

(2) closure transport responds constitutively to that structure,

(3) the induced response tensor is anisotropic,

(4) the anisotropic response varies directionally in a non-integrable way,

(5) the resulting Levi-Civita contraction is not everywhere degenerate.

Then the late-time closure-helicity pseudoscalar

$$H_{\text{late}} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} u_\mu J_\nu \Omega_{\rho\sigma}$$

is nonzero on a nonempty subset of D.

Proof. By the late-time generation theorem, direction-dependent support induces a nontrivial transverse sector $Q_\mu \neq 0$. By the anisotropic non-integrability lemma, this yields $\Xi_{\mu\nu} \neq 0$. The existence theorem for nonzero closure helicity from section 11 then implies $H_{\text{late}} \neq 0$ on a nonempty subset.

13 Non-cancellation conditions for the line-of-sight integral

The previous section showed that nonzero late-time closure helicity can be generated locally. That is still not enough for observation. The observed field is

$$\alpha_{\text{late}}(\hat{n}) = \int_{\gamma(\hat{n})} K(x) H_{\text{late}}(x)\, d\lambda,$$

where

$$K(x) := W(x) A_{\text{loc}}(x) \chi(\Phi(x)).$$

So one must control cancellation.

13.1 Path decomposition

Definition 13.1 (Admissible path partition)**.** Let $\gamma(\hat{n})$ be an admissible line of sight. An admissible path partition is a decomposition

$$\gamma(\hat{n}) = \bigcup_i I_i$$

into contiguous segments I_i on which the sign and magnitude structure of KH_{late} may be treated piecewise.

Definition 13.2 (Signed segment contribution)**.** For each segment I_i, define

$$C_i(\hat{n}) := \int_{I_i} K(x) H_{\text{late}}(x)\, d\lambda.$$

Then

$$\alpha_{\text{late}}(\hat{n}) = \sum_i C_i(\hat{n}).$$

Definition 13.3 (Dominant coherent sector)**.** A line of sight contains a dominant coherent sector if there exists a subcollection of path segments such that

(1) all corresponding C_i have the same sign,

(2) their total magnitude exceeds the total magnitude of all opposite-sign contributions,

(3) the corresponding sectors are admissible and non-negligibly weighted by K.

13.2 Coherent-domain model

Definition 13.4 (Coherent closure-helicity domain)**.** A coherent closure-helicity domain is a connected post-threshold region D_a such that

(1) H_{late} has fixed sign on D_a,

(2) $K(x)$ is bounded below by a positive constant on a nontrivial subset of D_a,

(3) the line of sight intersects D_a over a segment of positive measure.

Definition 13.5 (Environment-class coherence)**.** A family of coherent domains is environment-class coherent if the sign of H_{late} is determined by a shared structural environment class rather than varying randomly from one domain to the next.

Lemma 13.6 (Segment-bias lemma)**.** *Let $\gamma(\hat{n})$ intersect a family of coherent closure-helicity domains. Suppose there exists a same-sign subfamily whose weighted total path measure exceeds that of all opposite-sign coherent domains along the path. Then*

$$\alpha_{\text{late}}(\hat{n}) \neq 0.$$

Proof. On each same-sign domain the integrand KH_{late} has fixed nonzero sign and positive weighted measure. Therefore each contribution has the same sign. By hypothesis the total magnitude of the same-sign contribution exceeds the total magnitude of all opposite-sign contributions. Hence the full line integral cannot cancel.

Theorem 13.7 (Late-time non-cancellation theorem)**.** *Let $\gamma(\hat{n})$ be an admissible late-time line of sight. Suppose*

(1) H_{late} is generated by environment-conditioned anisotropic closure transport,

(2) the resulting source decomposes into coherent closure-helicity domains,

(3) at least one environment class induces same-sign domains along a nonzero-measure set of sightlines,

(4) the admissibility weight K does not overwhelmingly suppress those domains.

Then there exists a nonzero-measure set of sky directions $\mathcal{U} \subset S^2$ such that

$$\alpha_{\text{late}}(\hat{n}) \neq 0 \qquad \textit{for all } \hat{n} \in \mathcal{U}.$$

Proof. By hypothesis, the source decomposes into coherent sign domains. Since one environment class induces same-sign domains along a nonzero-measure set of sightlines and K does not overwhelmingly suppress them, those same-sign contributions furnish a dominant coherent sector on each such line of sight. The segment-bias lemma then implies the line-of-sight integral is nonzero on that set of directions.

Corollary 13.8 (Environment dependence)**.** *If the distribution of coherent closure-helicity domains varies across lines of sight, then the resulting late-time birefringence field is direction-dependent:*

$$\alpha_{\text{late}}(\hat{n}_1) \neq \alpha_{\text{late}}(\hat{n}_2)$$

for generic distinct directions $\hat{n}_1, \hat{n}_2$.

Proof. Different directions sample different weighted mixtures of coherent domains. Unless that distribution is exactly sky-uniform, the resulting integrals differ.

Proposition 13.9 (Exact cancellation requires tuning)**.** *Within the late-time branch, exact line-of-sight cancellation of KH_{late} on every admissible direction requires a tuned balance between positive and negative coherent-domain contributions across all paths.*

Proof sketch. Exact cancellation on every direction imposes one functional constraint per line of sight on the signed weighted domain distribution. Since the late-time branch allows anisotropic domain structure and direction-dependent support, these constraints overdetermine the admissible configuration space. Hence exact cancellation is a tuned rather than natural case.

14 Environment-class coherence theorem

The results of the present section are conditional on a constitutive regularity assumption: structural continuity of the environment-to-response map within an admissible class. This assumption is not yet derived from a deeper closure-dynamics law in the present paper and should be read accordingly. The purpose of the section is to establish what follows if such admissible constitutive continuity holds.

The previous section assumes coherent sign domains. We now prove that late-time environment classes naturally generate such domains under that regularity assumption.

14.1 Environment classes and response continuity

Definition 14.1 (Late-time environment class)**.** A late-time environment class $\mathcal{E}$ is a connected equivalence class of post-threshold regions whose admissibility-weighted transport structure agrees up to bounded perturbation in:

(1) directional support anisotropy,

(2) closure-memory skew,

(3) line-of-sight asymmetry,

(4) the effective anisotropic closure-response tensor ${\mathcal{A}_\mu}^\nu$.

Definition 14.2 (Environment-to-response map)**.** Let

$$\mathfrak{T} : \mathcal{E} \mapsto {\mathcal{A}_\mu}^\nu$$

be the map from environment class to anisotropic closure-response tensor.

Principle 14.3 (Structural continuity of admissible response)**.** Within an admissible late-time environment class, the induced anisotropic closure-response tensor ${\mathcal{A}_\mu}^\nu$ varies continuously under bounded perturbations of the defining structural variables.

Lemma 14.4 (No arbitrary sign flicker in admissible classes)**.** *Suppose $\mathfrak{T}$ is structurally continuous on an admissible environment class $\mathcal{E}$. Then the sign of H_{late} cannot flip arbitrarily densely throughout $\mathcal{E}$ unless $\mathcal{E}$ lies on a sign-degeneracy boundary where $H_{\text{late}} = 0$.*

Proof. H_{late} is built continuously from u^μ, J^μ, and $\Xi_{\rho\sigma}$, which depend continuously on ${\mathcal{A}_\mu}^\nu$ and Φ under the continuity hypothesis. A continuous scalar field cannot change sign arbitrarily densely on a connected class unless it repeatedly crosses zero. Therefore dense arbitrary sign flicker requires the class to lie on a set where $H_{\text{late}} = 0$.

14.2 Sign-degeneracy boundaries

Definition 14.5 (Sign-degeneracy boundary)**.** A sign-degeneracy boundary is a subset of the late-time environment space on which

$$H_{\text{late}} = 0$$

because the Levi-Civita contraction becomes degenerate, or because the anisotropic transport sector vanishes, or because antisymmetric transport content collapses.

Proposition 14.6 (Degeneracy is a restricted condition)**.** *The condition*

$$H_{\text{late}} = 0$$

imposes a nontrivial algebraic or differential relation among u^μ, J^μ, and $\Xi_{\rho\sigma}$. Therefore the sign-degeneracy set is a restricted subset of the admissible anisotropic transport configuration space.

Proof. The vanishing of

$$\epsilon^{\mu\nu\rho\sigma} u_\mu J_\nu \Xi_{\rho\sigma}$$

is a scalar constraint on the constitutive data. Unless one of the sectors vanishes identically by law, this is not automatically satisfied. Therefore it cuts out a restricted subset of the admissible configuration space.

14.3 Environment-class coherence

Theorem 14.7 (Environment-class coherence theorem)**.** *Let $\mathcal{E}$ be a connected admissible late-time environment class. Suppose*

(1) the induced anisotropic closure-response tensor $\mathcal{A}_\mu{}^\nu$ depends continuously on the defining structural variables of $\mathcal{E}$,

(2) the resulting transverse transport sector

$$Q_\mu = \mathcal{A}_\mu{}^\nu \nabla_\nu \Phi$$

is nontrivial on $\mathcal{E}$,

(3) the antisymmetric transport content $\Xi_{\rho\sigma}$ is nontrivial on $\mathcal{E}$,

(4) $\mathcal{E}$ is not contained in the sign-degeneracy boundary $H_{\text{late}} = 0$.

Then $\mathcal{E}$ contains an open or positive-measure subset on which

$$\operatorname{sgn}(H_{\text{late}})$$

is constant.

Proof. By continuity, H_{late} is continuous on $\mathcal{E}$. Since $\mathcal{E}$ is not contained in the degeneracy boundary, there exists at least one point where $H_{\text{late}} \neq 0$. By continuity there is a neighbourhood of that point on which H_{late} retains the same sign. This neighbourhood is an open or positive-measure subset of $\mathcal{E}$.

Definition 14.8 (Sign-connected environment class)**.** A connected admissible late-time environment class $\mathcal{E}$ is sign-connected if no path within $\mathcal{E}$ crosses the degeneracy boundary $H_{\text{late}} = 0$.

Corollary 14.9 (Class-wide sign coherence)**.** *If $\mathcal{E}$ is sign-connected, then*

$$\operatorname{sgn}(H_{\text{late}})$$

is constant on all of $\mathcal{E}$.

Proof. A continuous scalar on a connected set can change sign only by crossing zero. By sign-connectedness, that cannot occur within $\mathcal{E}$.

Corollary 14.10 (Coherence-to-non-cancellation)**.** *Suppose a line of sight $\gamma(\hat{n})$ intersects a family of admissible sign-connected late-time environment classes, and suppose one sign dominates in weighted path measure. Then*

$$\alpha_{\text{late}}(\hat{n}) \neq 0.$$

Proof. Each sign-connected class contributes a segment of definite helicity sign. If one sign dominates in weighted measure, the segment-bias lemma applies and the line integral is nonzero.

15 Scope, failure modes, and exact theorem status

The paper does not claim a full proof that cosmic birefringence is a DT phenomenon. It does something narrower and more precise.

15.1 What has been established

The following items have been established at theorem-architecture level:

(1) a DT-native definition of cosmic birefringence;

(2) a constitutive parity-sensitive closure field

$$\Pi = A_{\text{loc}}\chi(\Phi)H_{\text{cl}};$$

(3) a line-of-sight transport functional

$$\alpha_{\text{DT}}(\hat{n}) = \int_{\gamma(\hat{n})} W\Pi\, d\lambda;$$

(4) a formal decomposition into late-time and fossil branches;

(5) an observational proxy and null architecture;

(6) branch-specific proof obligations;

(7) existence conditions for nonzero closure helicity;

(8) a constructive theorem chain generating late-time closure helicity from anisotropic support and anisotropic response;

(9) a non-cancellation theorem for the late-time branch;

(10) an environment-class coherence theorem yielding same-sign helicity domains.

15.2 What remains conditional

The following items remain open:

(1) a deeper derivation of structural continuity of the environment-to-response map from first admissibility principles;

(2) a tighter genericity theorem for the distribution of sign-connected classes in realistic cosmic environments;

(3) the fossil-branch proof chain, including activation asymmetry and imprint persistence;

(4) full data-facing implementation against actual birefringence reconstructions;

(5) translation of the exploratory map-space coherence anomaly into a genuine rotation-field statement or a constitutive late-time proxy statement.

15.3 Current exploratory data-facing status

The exploratory map-space implementation described in sections 9 and 16 yields a split but informative result.

1. A multiscale coarse-cell coherence asymmetry survives internal shuffle nulls at a modest but nontrivial level in the Commander product, remains stable under tail jackknife, and persists across coarse resolutions.

2. The preferred coherence-axis family is not Commander-specific. It is reproduced strongly by SMICA at the intermediate and finer tested coarse scales. The 143-GHz foreground-subtracted SEVEM product agrees with that family at the intermediate scale, but does not remain cleanly aligned at the finest tested scale.

3. The 100-GHz foreground-subtracted SEVEM product does not track that same axis family at the coarser and intermediate tested scales, indicating that the present anomaly candidate is not a generic all-frequency cleaned-map feature.

4. A signed local-angle transport construction does not yet survive shuffle nulls strongly enough to count as an observational success for the stronger transport claim.

Accordingly, the present paper now supports a narrower empirical statement than a completed birefringence interpretation. There exists a product-persistent anomaly candidate in masked large-angle polarisation coherence structure, with the strongest persistence presently carried by the Commander–SMICA pair and intermediate-scale corroboration from the 143-GHz foreground-subtracted SEVEM product. The stronger claim of a robust signed transport field remains open.

15.4 Failure modes

The framework fails or is significantly weakened if any of the following are true:

(1) admissible closure transport is always parity-neutral, so $H_{\mathrm{cl}} = 0$;

(2) parity-sensitive closure transport can be written formally but is always structurally inadmissible;

(3) $H_{\mathrm{late}} \neq 0$ only in sign-random form with overwhelming line-of-sight cancellation;

(4) signed structural proxies do not outperform parity-even scalar proxies;

(5) sign-scramble nulls do not materially reduce the association;

(6) the apparent signal is strongly frequency-unstable in a manner characteristic of ordinary contamination;

(7) neither late-time nor fossil proxy architectures outperform broad nuisance sky structure.

16 Test-ready observable outputs

The previous sections define a full constitutive and statistical architecture. Before moving to data-facing work, it is useful to compress that structure into a small number of directly testable outputs. This section states the minimum operational products implied by the present paper.

16.1 Primary dependent variable

The primary dependent variable for the full intended data-facing programme is an observed birefringence rotation field

$$\alpha_{\rm obs}(\hat{n}),$$

reconstructed from CMB polarisation data.

The present paper has not yet inserted such an external rotation-field product into the pipeline. Instead, the current revision records an earlier map-space stage in which the cleaned large-angle polarisation fields themselves were tested for persistent sign-preserving coherence structure. Accordingly, $\alpha_{\rm obs}(\hat{n})$ remains the correct target variable for programme completion, but not the variable used in the exploratory implementation reported here.

16.2 Primary proxy products

For the full rotation-field stage, the first testing round should use the following three proxy products.

(1) **Parity-even scalar environment proxy** $D(\hat{n})$, used only as a baseline comparator.

(2) **Late-time signed structural proxy**

$$P_{\rm late}(\hat{n}) = P^{\rm late}_{\rm act}(\hat{n})\, P^{\rm late}_{\rm adm}(\hat{n})\, P^{\rm late}_{\rm hand}(\hat{n}).$$

(3) **Fossil or large-angle parity proxy**

$$P_{\rm fossil}(\hat{n}) = P_{\rm prim}(\hat{n})\, P_{{\rm low}\ell}(\hat{n})\, P_{\rm par}(\hat{n}).$$

The first operational question is whether $P_{\rm late}$ outperforms D. The second is whether $P_{\rm fossil}$ retains explanatory power after late-time environment conditioning is introduced.

16.3 Primary statistics

For the full rotation-field stage, the first testing round should use a deliberately small set of statistics.

(i) Signed correlation or regression coefficient. At least one primary statistic must preserve sign information:

$$\mathrm{Perf}\big(\alpha_{\rm obs}, P_{\rm late}\big), \qquad \mathrm{Perf}\big(\alpha_{\rm obs}, P_{\rm fossil}\big).$$

(ii) Sector contrast. For a signed proxy P, define

$$\Delta_{\rm sec}(\alpha; P) = \langle\alpha\rangle_{S_+} - \langle\alpha\rangle_{S_-},$$

where S_+ and S_- are high-sign and low-sign sectors.

(iii) Environment-conditioning gain. Measure the gain when moving from an unconditioned comparison to an environment-conditioned proxy model.

16.4 Mandatory nulls

A full DT-based rotation-field test is not serious unless it survives the following nulls.

(1) **Sign-destruction null:** preserve magnitude and support while scrambling the sign of the handed proxy.

(2) **Environment-preserving handedness-destruction null:** preserve activation and admissibility structure while destroying coherent handed structure.

(3) **Sky-remapping null:** preserve broad field statistics while breaking true sky alignment.

(4) **Random-sector null:** compare observed sector contrasts against matched random sky sectors.

(5) **Frequency-stability check:** test whether the inferred relation is stable across observing bands after standard cleaning.

16.5 Minimal decision logic

The full rotation-field testing round should be interpreted according to the following rules.

(1) If signed structural proxies do not outperform parity-even scalar proxies, the late-time closure-helicity reading is weakened.

(2) If sign-destruction nulls do not materially degrade performance, the handed transport component is not load-bearing.

(3) If environment conditioning materially improves the late-time proxy model, the late-time branch is strengthened.

(4) If broad parity-sensitive large-angle proxies retain explanatory power while environment gain remains weak, the fossil branch is comparatively strengthened.

(5) If the apparent relation is strongly frequency-unstable in a foreground-like way, the structural-geometric interpretation is weakened.

16.6 Operational summary

The first full rotation-field objective is not a global claim of discovery. It is the narrower and more disciplined task of determining whether

$$\alpha_{\mathrm{obs}}(\hat{n})$$

tracks a sign-preserving structured proxy more naturally than it tracks parity-even scalar environment structure.

The current revision precedes that stage. What has already been completed is an exploratory map-space implementation testing whether the cleaned large-angle polarisation fields themselves contain a stable product-persistent coherence-axis structure worthy of further constitutive interpretation.

16.7 Current implementation note

The current implementation has only partially reached the full objective just stated. No external birefringence rotation map was yet inserted into the pipeline, and therefore the present empirical outputs remain one step earlier in the decision chain. What has been tested is whether the cleaned large-angle polarisation maps already exhibit a stable multiscale coherence-axis structure that is worthy of further constitutive interpretation.

The present revision records the following test-ready outputs from that first implementation.

(1) A masked Commander coarse-cell coherence field exhibits a preferred axis family that survives shuffle nulls at a modest level and remains stable under 1%–5% tail jackknife.

(2) The same axis family is reproduced strongly by SMICA at $N_{\mathrm{side}} = 64$ and 128, and is matched by the 143-GHz foreground-subtracted SEVEM product at $N_{\mathrm{side}} = 64$.

(3) The 100-GHz foreground-subtracted SEVEM product does not follow that same axis family, indicating that the current anomaly candidate is not a generic all-frequency cleaned-map feature.

(4) A signed local-angle transport field built from nearest-neighbour orientation differences does not survive internal shuffle nulls strongly enough to carry the explanatory burden of the stronger transport claim.

These outputs imply that the next full test should begin not from a claim of detection, but from a sharpened target: the constitutive programme must explain why a product-persistent coherence-axis family appears in Commander and SMICA and why that family is compatible with 143-like behaviour more readily than with the present 100-GHz foreground-subtracted SEVEM product.

17 Current exploratory map-space findings and consequences

This section records, in paper form, the exploratory numerical outputs that motivated the present rewrite. The purpose is not to inflate those outputs into claims they do not yet support. It is to place them explicitly inside the theorem programme so that the constitutive and observational burdens are no longer discussed in purely abstract form.

17.1 Commander coherence-axis result

Using the common polarisation mask, the Commander Q/U field was compressed into nested coarse cells and analysed through a local coherence statistic formed from the ratio of the magnitude of the complex cell mean to the mean local polarisation amplitude. The resulting coarse-cell coherence field showed a preferred hemispheric asymmetry axis. That axis survived a shuffle null at a modest but nontrivial level and remained stable under removal of the strongest 1%–5% of cells. The significance level is not yet strong enough to justify a discovery-style claim, but it is strong enough to reject the description “mere numerical mush” as an adequate summary of the map-space result.

17.2 Multiscale persistence

The most important feature of the coherence result is not the raw score in any one run, but the persistence of the preferred axis under change of coarse resolution. For the Commander map the preferred coherence-axis family remained in the same broad sky sector as the coarse-cell resolution was changed from $N_{\mathrm{side}} = 32$ to 64 to 128, and the asymmetry score increased rather than collapsing as the field was resolved more finely. This behaviour is exactly the opposite of what one expects from a purely unstable coarse-pixel artefact.

17.3 Cross-product persistence

The next decisive step was to ask whether the Commander axis family is product-specific. It is not. When the same pipeline was applied to SMICA, the preferred axis converged strongly with the Commander result at the two most informative tested scales. At $N_{\text{side}} = 64$ the Commander–SMICA angular separation was only a few degrees, and at $N_{\text{side}} = 128$ the two were nearly coincident. This is presently the strongest internal cross-check in the entire empirical chain.

The 143-GHz foreground-subtracted SEVEM product then supplied a narrower but still meaningful form of corroboration. At $N_{\text{side}} = 64$ its preferred axis lay close to the Commander–SMICA family, strengthening the case that the anomaly candidate is compatible with a 143-like product. However, that agreement did not remain clean at $N_{\text{side}} = 128$. The 100-GHz foreground-subtracted SEVEM product behaved differently again and did not track the Commander–SMICA axis family at the coarser and intermediate tested scales. The current cross-product structure therefore points not to universal persistence across all cleaned products, but to a more selective persistence whose strongest support is carried by the Commander–SMICA pair.

17.4 What the cross-product pattern means

The cleanest reading of the present map-space evidence is therefore selective persistence rather than universal persistence. There exists a preferred coherence-axis family near roughly (155°–165°, 48°–50°) whose strongest support is carried by Commander and SMICA, and which is additionally matched by the 143-GHz foreground-subtracted SEVEM product at the intermediate tested scale. That family is not reproduced cleanly by the 100-GHz foreground-subtracted SEVEM product. Hence the present anomaly candidate is neither a Commander-only curiosity nor a generic all-product, all-frequency fact.

17.5 Failure of the stronger signed-transport claim

A more ambitious local signed-transport field was then constructed from nearest-neighbour angle differences. After imposing stricter neighbour-support requirements, that transport field no longer reduced to one-neighbour instabilities; however, its best asymmetry score did not survive internal shuffle nulls strongly enough to carry real constitutive weight. This is important. It means the exploratory work strengthens the weaker claim of a persistent coherence-axis anomaly candidate, but it does *not* yet strengthen the stronger claim that a bona fide signed local transport field has already emerged from the present data products.

17.6 Consequences for the theorem programme

These findings alter the burden structure of the paper in a productive way.

1. The constitutive programme no longer faces a vacuum of data-facing structure. It now faces a specific product-persistent coherence-axis candidate that needs explanation.

2. The stronger signed-transport proposal remains theoretically motivated but observationally under-supported by the current implementation.

3. The most natural next data-facing move is no longer a generic search over arbitrary proxies. It is now a focused requirement of the next stage to determine whether the Commander–SMICA–143 family is reproduced in a true rotation-field analysis or in a sharper DT late-time proxy construction with sign-preserving environment structure.

The correct summary is therefore exact: the exploratory map-space results strengthen the motivation for the DT birefringence programme, narrow the observational target, and simultaneously rule out any temptation to overstate what has already been shown.

18 Conclusion

This paper has developed a full theorem architecture for cosmic birefringence inside Darke Theory. The central move is to treat the phenomenon not as an externally appended parity-violating interaction in an otherwise neutral geometric background, but as a possible observational trace of closure-conditioned propagation in an emergent post-threshold regime.

The load-bearing constitutive source is the admissibility-filtered closure-helicity pseudoscalar,

$$\Pi = A_{\mathrm{loc}}\chi(\Phi)H_{\mathrm{cl}},$$

with observed sky rotation field

$$\alpha_{\mathrm{DT}}(\hat{n}) = \int_{\gamma(\hat{n})} W(x)\Pi(x)\, d\lambda.$$

The paper has shown that pure scalar closure-density transport cannot generate birefringence, that direction-dependent late-time support induces anisotropic closure response, that anisotropic response generates transverse closure transport, that non-integrable anisotropic response yields nonzero local late-time closure helicity, and that under environment-class coherence the corresponding line-of-sight integral is not naturally driven to exact cancellation. This yields the first constructive theorem chain within the late-time branch.

The fossil branch remains less complete, but its role has been formalised and its proof obligations stated clearly. The observational proxy architecture and null structure are also now sharply defined. The theory is therefore organised at the level of a constitutive theorem programme rather than a loose suggestion.

The central unresolved question is now equally sharp:

> Can nonzero closure helicity be shown to arise admissibly and persist nontrivially across cosmological propagation in the actual universe?

The present paper does not yet answer that question completely. It does something narrower but necessary. It identifies the exact constitutive object, the exact branch structure, the exact observational discriminants, and the exact proof hinges on which the answer depends.

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